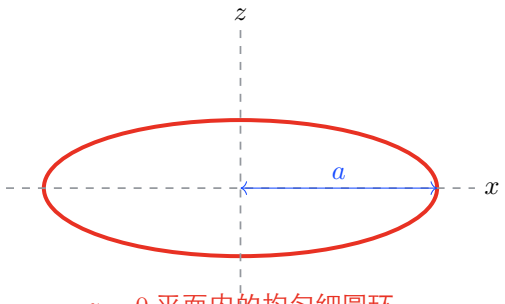
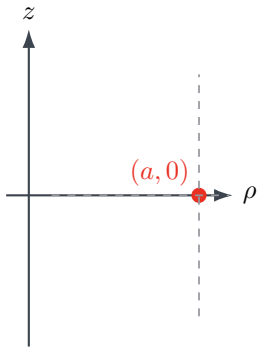
细圆环的体密度：两个 δ 锁定子午截面点，方位角仍自由



$z=0$ 平面内的均匀细圆环



子午截面中的支撑

$$\rho_V = \frac{Q}{2\pi a}\delta(\rho - a)\delta(z)$$

$$dV = \rho\, d\rho\, d\phi\, dz$$

$$\int \rho_V dV = \frac{Q}{2\pi a} \cdot a \cdot 2\pi = Q$$